Supplementary Appendix for

Gender Differences in How Scientists Present the Importance of Their Research

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Supplementary Data and Methods

Data

We merged several databases to analyze potential gender differences in positive presentation of clinical research articles and life science articles more broadly. First, we extracted information from the *MEDLINE PubMed XML* database from 2002 to 2017 to serve as the core of our data. The data is maintained by the U.S. National Library of Medicine (NLM) which offers access and documentation on their website (e.g., https://www.nlm.nih.gov/databases/download/pubmed_medline.html). To selectively expand on data from the XML version, we also created a data crawler of the PubMed online interface using the BeautifulSoup package in python (for further documentation, see https://www.crummy.com/software/BeautifulSoup/bs4/doc/). In particular, this crawling technique increased our access to structured abstracts for clinical research articles. We started our analysis in 2002 since this was the year *PubMed* started to systematically record forenames of authors, which enabled our designation of probable author gender on a large scale.

We determined the probable gender of the authors through the *Genderize* database that allows gender assignment for a large number of authors.[1] At the time of initial submission, *Genderize* included 86,710 distinct forenames drawn from 74 countries and 81 languages. Recent tests of the accuracy and comprehensiveness of four gender assignment algorithms, using a control sample of gender-matched forenames from a US government office, found that *Genderize* provided the most accurate estimates of gender.[2] Our underlying code for calling the *Genderize* database with a large set of forenames has been posted to Figshare.[3] *Genderize* uses a variety of information, such as social media records, to assign a probability that an individual with a particular forename is a man or a woman. For example, *Genderize* designates the forename "Chris" as male with 93% probability based on 8,631 verified records in the database. For our analysis, we considered gender determined if *Genderize* assigned at least a 90% probability to its gender designation. Changing this probability threshold changed neither the gender distribution of lead authors nor our estimates of gender differences in the positive presentation of articles. Moreover, gender designation proved accurate for authors of different ethnic backgrounds (described below).

We also retrieved the journal impact factors (JIFs) for the journals publishing the research from the 2017 Journal Citation Report (Thomson Reuters). We tested bivariate correlations of various journal impact factors across the years included in our analyses and obtained correlation coefficients in excess of 0.90, indicating little temporal variance in the scaling of the metric. We therefore used the most recent JIFs for all years. We assigned journals without a listed impact to the lowest impact category, as the Journal Citation Report requires a minimum impact threshold for inclusion in the index.

Methods

Natural Language Processing

We used Natural Language Processing to identify the nouns that immediately follow the 25 positive words (adjectives). This analysis informs, for example, to what extent adjectives may be used in a more factual context (e.g., robust errors) versus qualifying the presented research (e.g., 'novel mechanism' or 'unique approach'). In brief, we used the POS (Part-of-Speech)-tagger implemented in the NLTK platform for python to tag all nouns following the positive word, keep the first identified noun, and apply lemmatization to group nouns of same origin (e.g., strategies and strategy are combined into strategy). Further details on this approach are documented online (https://www.nltk.org/).

Estimation

To assess whether an article presented the work positively, we used a set of 25 words that prior research identified as positive words that were frequently used in life science articles.[4]

We used linear probability models (LPM) to estimate relative probabilities with which lead authors of different gender use at least one of the 25 positive words, adjusting for journal fixed effects (based on journals' unique ISSN numbers), year of publication, and potential differences in the positive presentation of research findings across fields. Logistic regression as an alternative estimation model has two disadvantages in our analysis. First, the large number of intercepts, year and journal fixed effects (> 10,000), raise the possibility of incidental parameters bias and, in our case, prevented the convergence of some of our models for the broader PubMed database. Second, logistic regressions can overestimate effect sizes as a result of the high leverage of marginal cases (i.e., identifying larger gender differences than reported in the main text), whereas linear probability models average across observations and produce more conservative results.

To account for potential differences in positive presentation across research fields, we created a fine-grained control variable based on article keywords. Core ideas and concepts in the life sciences are identified by Medical Subject Headers (MeSH) terms that accompany all publications. The MeSH lexicon is a controlled vocabulary used by the U.S. National Library of Medicine to index articles for *PubMed*. Note, MeSH keywords are assigned not by authors but by professional science librarians trained specifically to perform this task, eliminating any bias that might arise from scientists being able to influence this keyword assignment.

We created our control variable for field of study by first calculating the proportion of all articles using a MeSH term that had one of the 25 positive terms in its title or abstract. To norm the scaling of this variable, we created a percentile ranking for each MeSH term. Since many articles are associated with more than one MeSH term, we used the median percentile ranking across all MeSH terms associated with an individual article. We refer to this variable as "percentile positive MeSH field" in our supplementary tables.

We also created several variables to assess how the composition of the author byline beyond the first and last authors' gender may affect the positive framing of an article. To test whether the gender composition of interior authors added explanatory power, we added a variable for the percentage of women on the author byline. As projects with more authors might have greater scale, we also included a control for the total number of authors.

Table S1. Clinical journals analyzed, ranked by journal impact factor

Clinical Journals	ISSN	Journal Impact Factor (JIF)
New England Journal Of Medicine	0028-4793	79.26
Lancet	0140-6736	53.25
Jama-Journal Of The American Medical Association	0098-7484	47.66
Lancet Oncology	1470-2045	36.42
Lancet Neurology	1474-4422	27.14
Journal Of Clinical Oncology	0732-183X	26.36
Lancet Infectious Diseases	1473-3099	25.15
Bmj-British Medical Journal	1756-1833	23.56
Lancet Respiratory Medicine	2213-2600	21.47
Jama Internal Medicine	2168-6106	19.99
Annals Of Internal Medicine	0003-4819	19.38
Lancet Diabetes & Endocrinology	2213-8587	19.31
Circulation	0009-7322	18.88
Lancet Global Health	2214-109X	18.71
Journal Of The American College Of Cardiology	0735-1097	16.83
Lancet Psychiatry	2215-0374	15.23
Diabetes Care	0149-5992	13.40
Journal Of Allergy And Clinical Immunology	0091-6749	13.26
Annals Of The Rheumatic Diseases	0003-4967	12.35
Lancet Hiv	2352-3018	11.36
Jnci-Journal Of The National Cancer Institute	0027-8874	11.24
Brain	0006-8950	10.85
Lancet Haematology	2352-3026	10.70
Clinical Cancer Research	1078-0432	10.20
Jacc-Cardiovascular Interventions	1936-8798	9.88
Clinical Infectious Diseases	1058-4838	9.12
Jama Dermatology	2168-6068	8.11
Clinical Gastroenterology And Hepatology	1542-3565	7.68
Mayo Clinic Proceedings	0025-6196	7.20
European Journal Of Cancer	0959-8049	7.19
Canadian Medical Association Journal	0820-3946	6.82
British Journal Of Anaesthesia	0007-0912	6.50
Journal Of The National Comprehensive Cancer Network	1540-1405	6.47
Stroke	0039-2499	6.24
British Journal Of Dermatology	0007-0963	6.13
British Journal Of Cancer	0007-0920	5.92
Clinical Journal Of The American Society Of Nephrology	1555-9041	5.84
Journal Of Clinical Endocrinology & Metabolism	0021-972X	5.79
Osteoarthritis And Cartilage	1063-4584	5.45
Clinical Science	0143-5221	5.22
Annals Of Family Medicine	1544-1709	4.54
Clinical Lung Cancer	1525-7304	4.20
Journal Of Pediatrics	0022-3476	3.67
Journal Of Rheumatology	0315-162X	3.47
European Journal Of Internal Medicine	0953-6205	3.28
Clinical Journal Of Pain	0749-8047	3.21
Journal Of Clinical Psychopharmacology	0271-0749	3.13
Annals Of Medicine	0785-3890	3.01
Journal Of Clinical Gastroenterology	0192-0790	2.97

Note: Print and electronic ISSN numbers were used for the identification of articles.

Table S2. Sample derivation for clinical articles and articles in broader PubMed

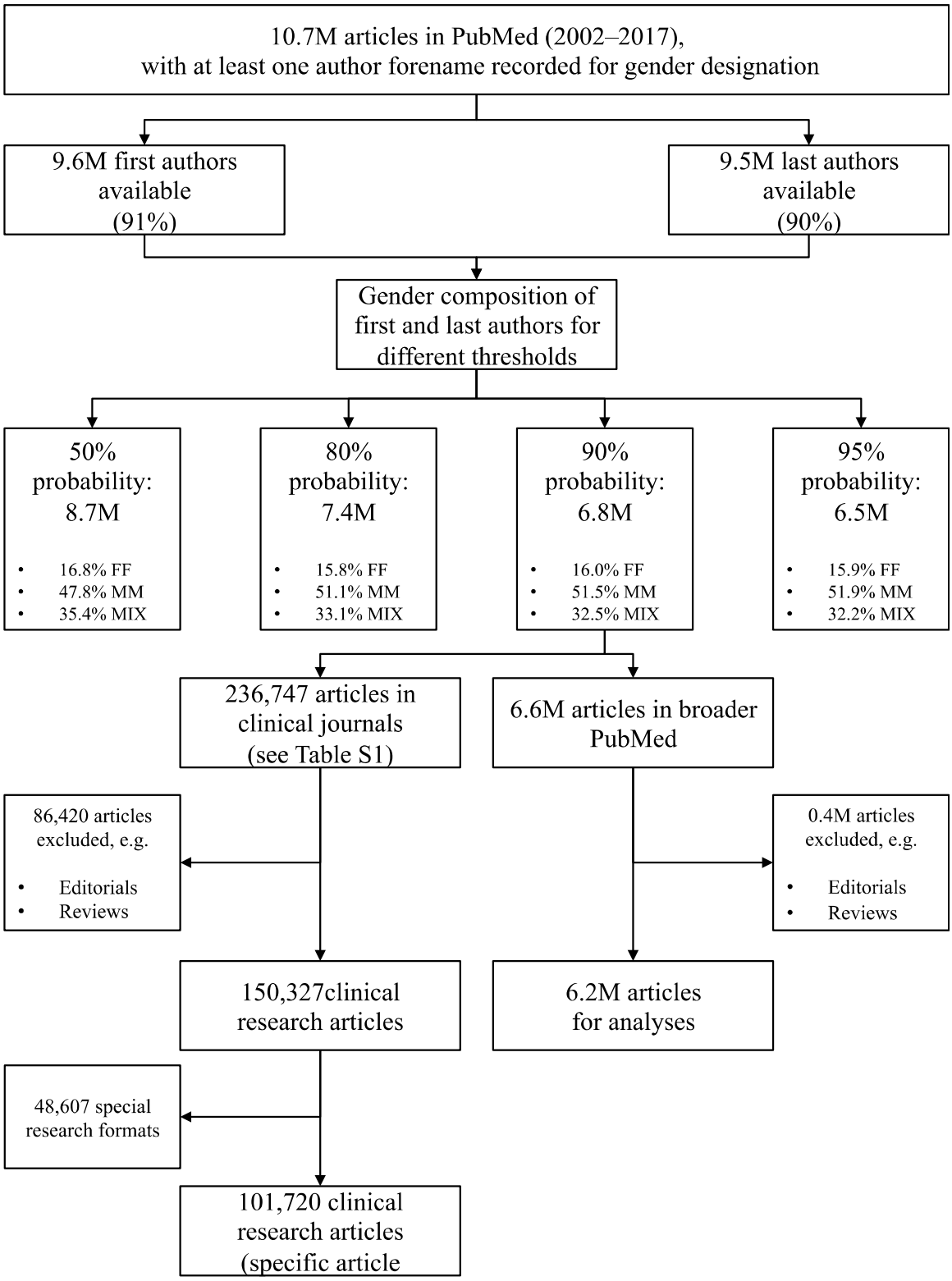


Table S3. Types of articles in clinical journals

Type of publication	Frequency	Rel. Frequency	Cumulative Frequency
Journal Article	**101,720**	**43.0**	**43.0**
Article in Press	16	0.0	43.0
Book	2	0.0	43.0
Conference Paper	2,421	1.0	44.0
Editorial	16,519	7.0	51.0
Erratum	43	0.0	51.0
Letter	39,539	16.7	67.7
NA	18,652	7.9	75.6
Note	33,170	14.0	89.6
Review	18,034	7.6	97.2
Short Survey	6,631	2.8	100.0
Total	236,747	100.0	

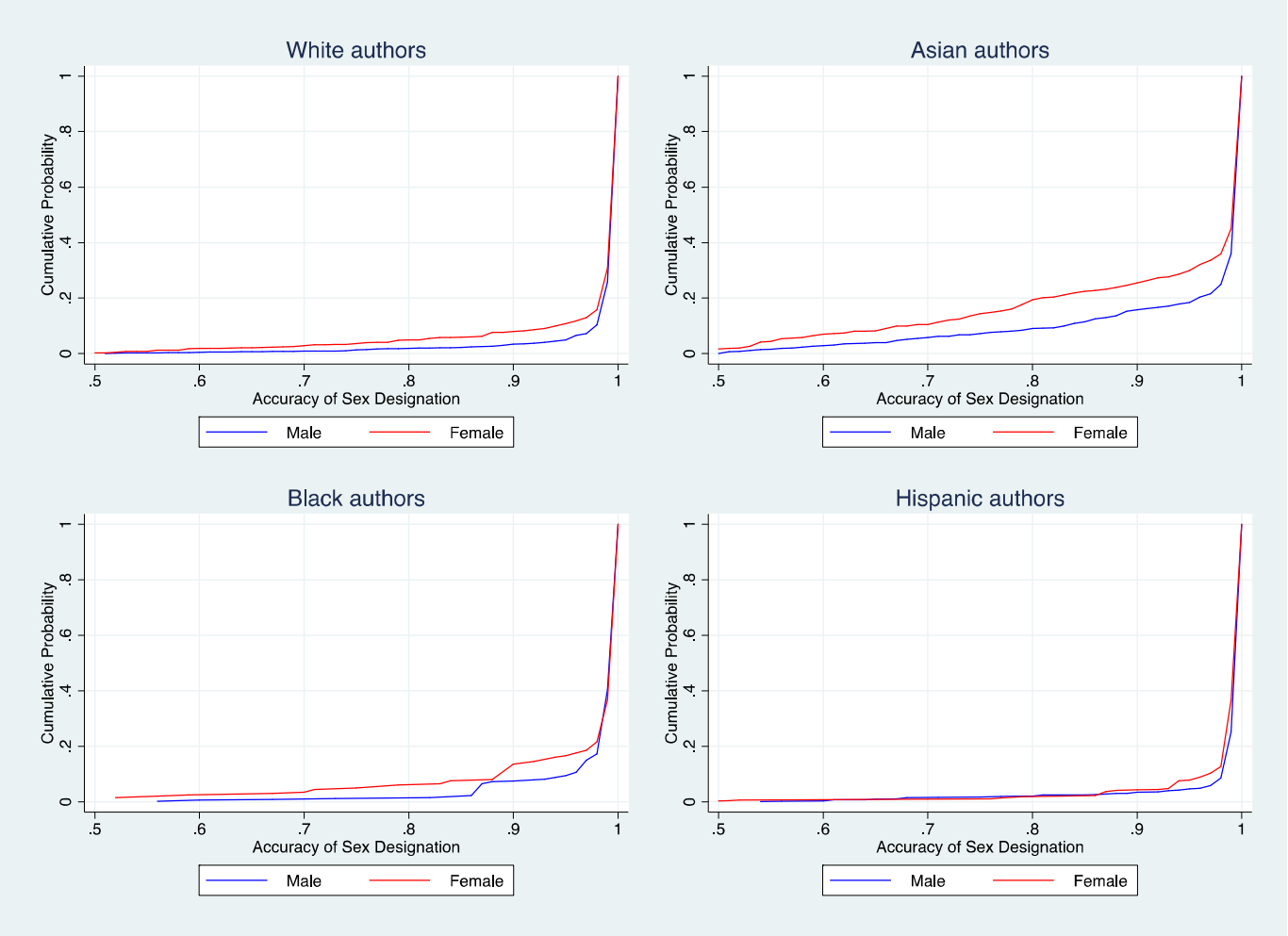
Note: Table reflects all articles appearing in the considered journals (Table S1) where the gender of the first and last author could be designated with at least 90% probability. Blue indicates special research formats and bold blue represents the core sample for analyses of clinical research articles.

Table S4. Types of articles in broader PubMed

Type of publication	Frequency	Rel. Frequency	Cumulative Frequency
Case Reports	346,510	5.6	5.6
Clinical Trial	79,178	1.3	6.8
Comment	248,006	4.0	10.8
Comparative Study	363,778	5.8	16.6
English Abstract	86,026	1.4	18.0
Evaluation Studies	73,081	1.2	19.2
Journal Article	4,959,559	79.4	98.6
Letter	89,974	1.4	100.0
Total	6,246,112	100.0	

Note: Table reflects publication types for articles analyzed in broader PubMed sample.

Figure S5. Accuracy of genderize algorithm for different ethnic backgrounds.



Note: The average probabilities with which gender could be designated were 98.56% for White authors (N = 44,623), 94.13% for Asian authors (N= 7,624), 97.52% for Black authors (N = 410), and 99.02% for Hispanic authors (N = 1,448).

Figure S6. Illustration of U.S. Census data for probabilistic designation of ethnicity

name	count	pctwhite	pctblack	pctapi	pctasian	pct2race	pcthispanic
SMITH	2,376,206	73.35	22.22	0.4	0.85	1.63	1.56
JOHNSON	1,857,160	61.55	33.8	0.42	0.91	1.82	1.5
WILLIAMS	1,534,042	48.52	46.72	0.37	0.78	2.01	1.6
BROWN	1,380,145	60.71	34.54	0.41	0.83	1.86	1.64
JONES	1,362,755	57.69	37.73	0.35	0.94	1.85	1.44
MILLER	1,127,803	85.81	10.41	0.42	0.63	1.31	1.43
DAVIS	1,072,335	64.73	30.77	0.4	0.79	1.73	1.58
GARCIA	858,289	6.17	0.49	1.43	0.58	0.51	90.81
RODRIGUEZ	804,240	5.52	0.54	0.58	0.24	0.41	92.7
WILSON	783,051	69.72	25.32	0.46	1.03	1.74	1.73
MARTINEZ	775,072	6.04	0.52	0.6	0.64	0.46	91.72
ANDERSON	762,394	77.6	18.06	0.48	0.7	1.59	1.58
TAYLOR	720,370	67.8	27.67	0.39	0.75	1.78	1.61
THOMAS	710,696	55.53	38.17	1.63	1.01	2	1.66
HERNANDEZ	706,372	4.55	0.38	0.65	0.27	0.35	93.81
MOORE	698,671	68.85	26.92	0.37	0.65	1.7	1.5
MARTIN	672,711	77.47	15.3	0.71	0.94	1.59	3.99
JACKSON	666,125	41.93	53.02	0.31	1.04	2.18	1.53
THOMPSON	644,368	72.48	22.53	0.44	1.15	1.78	1.62
WHITE	639,515	67.91	27.38	0.39	1.01	1.76	1.55
LOPEZ	621,536	5.85	0.61	1.04	0.47	0.52	91.51

Note: Ethnic groups included in U.S. Census data include White, Black, Asian Pacific Islander (API), Asian, self-identification as belonging to two ethnic groups (2race); Hispanic.

Table S7. Positive words (adjectives) and location in structured abstract, overall and by gender

All structured abstracts

Rank	Word	Background	Methods	Results	Conclusions	Total occurences
1	novel	35%	8%	14%	43%	4,756
2	favorable	13%	7%	41%	39%	1,722
3	promising	36%	1%	2%	61%	1,243
4	unique	24%	14%	28%	33%	960
5	excellent	9%	3%	49%	38%	825
6	robust	15%	6%	43%	36%	719
7	prominent	20%	5%	49%	25%	415
8	supportive	15%	28%	30%	27%	373
9	encouraging	15%	0%	5%	79%	272
10	remarkable	16%	2%	45%	37%	122
11	innovative	28%	9%	9%	53%	109
12	bright	20%	27%	48%	5%	66
13	unprecedented	56%	0%	15%	28%	39
14	reassuring	0%	0%	21%	79%	28
15	enormous	54%	0%	0%	46%	26
16	hopeful	11%	11%	67%	11%	9
17	creative	0%	38%	38%	25%	8
18	assuring	75%	0%	25%	0%	4
19	astonishing	67%	0%	33%	0%	3
20	inventive	0%	0%	100%	0%	1
21	spectacular	0%	0%	100%	0%	1
22	amazing	0%	0%	0%	0%	0
23	groundbreaking	0%	0%	0%	0%	0
24	inspiring	0%	0%	0%	0%	0
25	phenomenal	0%	0%	0%	0%	0
	Total	**26%**	**8%**	**24%**	**42%**	

Lead Authors	Background	Methods	Results	Conclusions	Total occurences
Female-female articles	**25%**	**8%**	**26%**	**41%**	1,239
Male-male articles	**26%**	**8%**	**24%**	**42%**	5,979
All but female-female articles	**26%**	**8%**	**24%**	**43%**	10,462

Legend	0-10%	11-20%	21-30%
	31-40%	41-50%	>50%

Note: Based on 62,877 structured abstracts (81% of available abstracts for clinical research articles)

Table S8. Positive words (adjectives) and associated nouns in structured abstracts

Novel		Unique		Favorable		Promising		Robust	
Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.
target	4%	patient	5%	outcome	19%	approach	8%	sensitivity	4%
approach	4%	opportunity	4%	prognosis	6%	strategy	7%	predictor	3%
mutation	4%	population	3%	effect	5%	target	6%	response	3%
strategy	4%	feature	2%	response	4%	activity	6%	association	3%
mechanism	3%	mechanism	2%	safety	3%	result	6%	evidence	3%
therapy	3%	study	1%	risk	3%	treatment	5%	method	2%
agent	2%	property	1%	factor	3%	tool	4%	effect	2%
marker	2%	gene	1%	toxicity	2%	candidate	3%	expression	2%
treatment	2%	mutation	1%	profile	2%	therapy	3%	adjustment	1%
method	2%	effect	1%	change	2%	efficacy	2%	range	1%

Table S9. Positive words (adjectives) and associated nouns in titles and abstracts of broader life science articles (PubMed)

Table S9a. Titles

Novel		Unique		Favorable		Promising		Robust	
Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.
mutation	3.73%	case	4.45%	outcome	20.49%	target	7.44%	method	3.75%
approach	3.70%	feature	2.71%	prognosis	10.45%	approach	6.14%	estimation	1.68%
method	2.19%	role	2.18%	response	5.10%	tool	4.52%	model	1.23%
target	1.58%	property	1.99%	effect	4.65%	strategy	3.56%	cell	1.07%
gene	1.38%	presentation	1.93%	risk	3.90%	treatment	2.93%	control	1.06%
technique	1.36%	mechanism	1.20%	factor	3.03%	drug	2.93%	approach	0.98%
mechanism	1.36%	gene	1.17%	term	2.47%	agent	2.73%	detection	0.91%
inhibitor	1.28%	challenge	1.07%	result	1.50%	therapy	2.64%	response	0.84%
strategy	1.06%	characteristic	1.01%	histology	1.20%	biomarker	2.17%	design	0.82%
role	1.04%	effect	0.88%	impact	1.12%	result	2.08%	gene	0.77%

Table S9b. Abstracts

Novel		Unique		Favorable		Promising		Robust	
Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.
approach	4.50%	property	3.85%	outcome	7.56%	approach	6.87%	method	3.34%
mechanism	2.81%	feature	3.38%	effect	4.58%	result	6.14%	response	2.33%
method	2.71%	opportunity	2.67%	prognosis	3.65%	candidate	5.30%	effect	1.37%
strategy	2.51%	challenge	1.85%	condition	2.97%	strategy	4.99%	model	1.33%
target	2.47%	case	1.53%	result	2.52%	target	4.60%	evidence	1.31%
agent	1.57%	characteristic	1.52%	response	2.40%	tool	3.61%	cell	1.09%
specie	1.50%	role	1.46%	property	2.33%	treatment	2.36%	approach	1.02%
therapy	1.42%	structure	1.44%	interaction	2.08%	agent	2.31%	data	0.99%
role	1.41%	mechanism	1.26%	safety	1.92%	drug	2.26%	expression	0.96%
drug	1.37%	ability	1.03%	profile	1.30%	application	1.97%	increase	0.95%

Table S10. Positive words (adjectives) and associated nouns of broader life science articles (PubMed) by gender

Table S10a. Female first and last authors

Novel		Unique		Favorable		Promising		Robust	
Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.
approach	4.03%	opportunity	3.09%	outcome	6.96%	approach	6.59%	method	3.10%
mechanism	3.04%	property	2.94%	effect	5.26%	strategy	5.43%	response	2.77%
target	2.95%	feature	2.91%	prognosis	3.88%	candidate	5.33%	evidence	1.83%
strategy	2.71%	challenge	2.66%	condition	3.58%	result	4.94%	effect	1.64%
specie	2.03%	role	1.79%	attitude	2.11%	target	4.79%	cell	1.30%
method	1.98%	characteristic	1.61%	property	2.09%	tool	3.80%	model	1.10%
role	1.87%	need	1.39%	response	2.05%	agent	2.69%	increase	1.09%
mutation	1.78%	structure	1.27%	result	1.71%	drug	2.48%	association	0.97%
drug	1.47%	contribution	1.15%	environment	1.67%	treatment	2.42%	change	0.94%
therapy	1.44%	mechanism	1.13%	safety	1.64%	application	2.06%	data	0.92%

Table S10b. Male first and last authors

Novel		Unique		Favorable		Promising		Robust	
Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.	Noun	Rel. Freq.
approach	4.94%	property	4.06%	outcome	8.16%	result	6.97%	method	3.32%
method	3.11%	feature	3.58%	effect	4.64%	approach	6.93%	response	1.97%
mechanism	2.53%	opportunity	2.61%	prognosis	3.42%	candidate	5.07%	model	1.34%
strategy	2.44%	challenge	1.84%	result	3.07%	strategy	4.56%	evidence	1.24%
target	2.18%	case	1.78%	condition	2.67%	target	4.15%	effect	1.14%
agent	1.74%	characteristic	1.51%	response	2.50%	tool	3.45%	approach	1.12%
therapy	1.51%	structure	1.46%	property	2.40%	treatment	2.38%	data	1.05%
technique	1.44%	mechanism	1.30%	safety	2.25%	agent	2.14%	cell	0.97%
drug	1.40%	role	1.26%	interaction	2.14%	drug	2.14%	expression	0.87%
treatment	1.23%	ability	1.05%	profile	1.31%	method	1.86%	increase	0.84%

Table S11. Linear probability model of probability of positive presentation

	(1)	(2)	(3)	(4)	(5)	(6)
	Clinical articles (all JIFs)	**Clinical articles (JIF >10)**	**Clinical articles (JIF ≤10)**	**PubMed articles (all JIFs)**	**PubMed articles (JIF >10)**	**PubMed articles (JIF ≤10)**
Adjusted proportion positive articles						
Female first and last author = 1	10.9%	10.7%	11.0%	11.4%	12.8%	11.3%
Female first and last author = 0	12.2%	12.9%	11.2%	11.5%	14.4%	11.3%
Absolute difference	**-1.3%**	**-2.3%**	**-0.2%**	**-0.1%**	**-1.6%**	**0.0%**
Relative difference	12.3%	21.4%	1.4%	0.6%	12.8%	0.1%
Linear probability model						
Dependent variable: Positive article = 1						
Independent variables						
Female first and last author = 1	**-0.013*****	**-0.023*****	**-0.002**	**-0.001**	**-0.016*****	**-0.000**
	[0.004]	[0.005]	[0.006]	[0.001]	[0.003]	[0.001]
Percent female authors on byline	-0.006	0.002	-0.014*	-0.010***	0.001	-0.011***
	[0.005]	[0.006]	[0.007]	[0.001]	[0.003]	[0.001]
Count of authors on byline	0.004***	0.004***	0.003***	0.002***	0.002***	0.002***
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Percentile positive MeSH field	0.003***	0.003***	0.004***	0.002***	0.002***	0.002***
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Publication year fixed effects						
Fixed_effect_2003	-0.005	-0.000	-0.014	0.003***	0.004	0.003***
	[0.006]	[0.007]	[0.010]	[0.001]	[0.004]	[0.001]
Fixed_effect_2004	-0.000	0.002	-0.004	0.006***	0.005	0.006***
	[0.006]	[0.007]	[0.010]	[0.001]	[0.004]	[0.001]
Fixed_effect_2005	0.003	0.004	0.001	0.010***	0.019***	0.009***
	[0.006]	[0.007]	[0.010]	[0.001]	[0.004]	[0.001]
Fixed_effect_2006	0.011*	0.013*	0.009	0.012***	0.014***	0.011***
	[0.006]	[0.007]	[0.010]	[0.001]	[0.004]	[0.001]
Fixed_effect_2007	0.019***	0.019***	0.022**	0.014***	0.006	0.015***
	[0.006]	[0.007]	[0.010]	[0.001]	[0.004]	[0.001]
Fixed_effect_2008	0.021***	0.024***	0.018*	0.014***	0.018***	0.013***
	[0.006]	[0.008]	[0.010]	[0.001]	[0.004]	[0.001]
Fixed_effect_2009	0.034***	0.039***	0.026***	0.013***	0.016***	0.013***
	[0.006]	[0.008]	[0.010]	[0.001]	[0.005]	[0.001]
Fixed_effect_2010	0.029***	0.036***	0.018*	0.014***	0.021***	0.014***
	[0.006]	[0.008]	[0.011]	[0.001]	[0.005]	[0.001]
Fixed_effect_2011	0.026***	0.033***	0.014	0.020***	0.019***	0.020***
	[0.006]	[0.008]	[0.010]	[0.001]	[0.005]	[0.001]
Fixed_effect_2012	0.033***	0.035***	0.029***	0.022***	0.015***	0.022***
	[0.006]	[0.008]	[0.010]	[0.001]	[0.005]	[0.001]
Fixed_effect_2013	0.045***	0.046***	0.043***	0.024***	0.021***	0.025***
	[0.006]	[0.008]	[0.010]	[0.001]	[0.005]	[0.001]
Fixed_effect_2014	0.039***	0.039***	0.038***	0.026***	0.030***	0.025***
	[0.006]	[0.008]	[0.010]	[0.001]	[0.005]	[0.001]
Fixed_effect_2015	0.053***	0.052***	0.053***	0.028***	0.036***	0.028***
	[0.006]	[0.008]	[0.010]	[0.001]	[0.005]	[0.001]
Fixed_effect_2016	0.049***	0.052***	0.045***	0.032***	0.047***	0.031***
	[0.006]	[0.008]	[0.010]	[0.001]	[0.005]	[0.001]
Fixed_effect_2017	0.059***	0.065***	0.053***	0.038***	0.064***	0.037***
	[0.007]	[0.010]	[0.012]	[0.001]	[0.006]	[0.001]
Constant	-0.045***	-0.040***	-0.054***	-0.026***	-0.015***	-0.026***
	[0.005]	[0.007]	[0.009]	[0.001]	[0.005]	[0.001]
Journal fixed effects	**YES**	**YES**	**YES**	**YES**	**YES**	**YES**
Observations	96,543	59,360	37,183	5,046,315	237,783	4,808,532
R-squared	0.084	0.102	0.050	0.094	0.082	0.094

Standard errors in brackets

*** p<0.01, ** p<0.05, * p<0.1

Note: Journal fixed effects included but not shown: 24 (Model 2); 25 (Model 3); 223 (Model 5); 9,503 (Model 6)

Table S12. Hierarchical LPM of positive presentation, clinical articles (JIF > 10)

Models	(1)	(2)	(3)	(4)	(5)	(6)
	Gender only coefficient	**Add publication year fixed effects**	**Add journal fixed effects**	**Add author byline features**	**Add positive control for field based on MeSH**	**Add abstract length (imputed)**
Dependent variable: Positive article = 1						
Independent variables						
Female first and last author = 1	**-0.066*** **	**-0.066*** **	**-0.029*** **	**-0.024*** **	**-0.023*** **	**-0.023*** **
	[0.004]	**[0.004]**	**[0.004]**	**[0.005]**	**[0.005]**	**[0.005]**
Percent female authors on byline				0.005	0.002	0.002
				[0.006]	[0.006]	[0.006]
Count of authors on byline				0.004***	0.004***	0.004***
				[0.000]	[0.000]	[0.000]
Percentile positive MeSH field					0.003***	0.003***
					[0.000]	[0.000]
Percentile abstract length						0.000
						[0.000]
Publication year fixed effects						
Fixed_effect_2003		-0.004	-0.002	-0.001	-0.000	-0.000
		[0.007]	[0.007]	[0.007]	[0.007]	[0.007]
Fixed_effect_2004		0.009	0.002	0.001	0.002	0.002
		[0.007]	[0.007]	[0.007]	[0.007]	[0.007]
Fixed_effect_2005		0.025***	0.007	0.006	0.004	0.004
		[0.007]	[0.007]	[0.007]	[0.007]	[0.007]
Fixed_effect_2006		0.017**	0.015*	0.014*	0.013*	0.013*
		[0.007]	[0.008]	[0.007]	[0.007]	[0.007]
Fixed_effect_2007		0.018**	0.017**	0.016**	0.019***	0.019***
		[0.007]	[0.008]	[0.007]	[0.007]	[0.007]
Fixed_effect_2008		0.040***	0.025***	0.022***	0.024***	0.024***
		[0.008]	[0.008]	[0.008]	[0.008]	[0.008]
Fixed_effect_2009		0.049***	0.041***	0.036***	0.039***	0.038***
		[0.007]	[0.008]	[0.008]	[0.008]	[0.008]
Fixed_effect_2010		0.046***	0.040***	0.034***	0.036***	0.036***
		[0.008]	[0.008]	[0.008]	[0.008]	[0.008]
Fixed_effect_2011		0.048***	0.037***	0.031***	0.033***	0.032***
		[0.008]	[0.008]	[0.008]	[0.008]	[0.008]
Fixed_effect_2012		0.050***	0.042***	0.033***	0.035***	0.035***
		[0.008]	[0.008]	[0.008]	[0.008]	[0.008]
Fixed_effect_2013		0.058***	0.053***	0.044***	0.046***	0.045***
		[0.008]	[0.008]	[0.008]	[0.008]	[0.008]
Fixed_effect_2014		0.049***	0.049***	0.037***	0.039***	0.038***
		[0.007]	[0.008]	[0.008]	[0.008]	[0.008]
Fixed_effect_2015		0.066***	0.064***	0.052***	0.052***	0.052***
		[0.007]	[0.008]	[0.008]	[0.008]	[0.008]
Fixed_effect_2016		0.072***	0.066***	0.051***	0.052***	0.052***
		[0.008]	[0.008]	[0.008]	[0.008]	[0.008]
Fixed_effect_2017		0.080***	0.069***	0.052***	0.065***	0.065***
		[0.007]	[0.008]	[0.008]	[0.010]	[0.010]
Constant	0.139***	0.102***	0.102***	0.073***	-0.040***	-0.042***
	[0.001]	[0.005]	[0.006]	[0.006]	[0.007]	[0.007]
Journal fixed effects	**NO**	**NO**	**YES**	**YES**	**YES**	**YES**
Observations	61,973	61,973	61,973	61,973	59,360	59,360
R-squared	0.005	0.011	0.082	0.090	0.102	0.102

Standard errors in brackets
*** p<0.01, ** p<0.05, * p<0.1

Note: Abstract length imputed for missing abstracts by drawing the percentile abstract length from a uniform distribution.

Table S13. Percent relative gender difference in positive framing clinical articles (JIF > 10), hierarchical LPM

	Model specification	Female first and last author	At least one male first or last author	Percent relative difference	N
(1)	Gender only coefficient	7.27%	13.89%	91.21%	61,973
(2)	Add publication year fixed effects	7.29%	13.89%	90.40%	61,973
(3)	Add journal fixed effects	10.38%	13.27%	27.92%	61,973
(4)	Add author byline features	10.75%	13.20%	22.72%	61,973
(5)	Add positive control for field based on MeSH	10.65%	12.93%	21.42%	59,360
(6)	Add abstract length imputed	10.66%	12.93%	21.30%	59,360

Table S14. LPM of positive presentation for clinical articles, gender–impact interaction

	Proportion positive articles	
Predicted margins	**Clinical articles (JIF >10)**	**Clinical articles (JIF ≤10)**
Female first and last author = 1	9.8%	10.5%
Female first and last author = 0	13.1%	11.3%
Linear Probability Model		
Dependent variable: Positive article = 1		
Independent variables	Pooled sample	
Female first and last author = 1	-0.007	
	[0.007]	
Publishing Journal JIF > 10	0.005	
	[0.014]	
JIF > 10 X Female first and last author	-0.025**	
	[0.010]	
Percent female authors on byline	-0.008	
	[0.010]	
Count of authors on byline	0.005***	
	[0.001]	
Percentile positive MeSH field	0.004***	
	[0.000]	
Publication year fixed effects		
Fixed_effect_2003	-0.007	
	[0.006]	
Fixed_effect_2004	-0.000	
	[0.006]	
Fixed_effect_2005	0.005	
	[0.007]	
Fixed_effect_2006	0.008	
	[0.010]	
Fixed_effect_2007	0.014	
	[0.010]	
Fixed_effect_2008	0.018*	
	[0.010]	
Fixed_effect_2009	0.029**	
	[0.011]	
Fixed_effect_2010	0.024**	
	[0.011]	
Fixed_effect_2011	0.024**	
	[0.010]	
Fixed_effect_2012	0.028**	
	[0.011]	
Fixed_effect_2013	0.040***	
	[0.012]	
Fixed_effect_2014	0.029**	
	[0.011]	
Fixed_effect_2015	0.044***	
	[0.015]	
Fixed_effect_2016	0.039***	
	[0.012]	
Fixed_effect_2017	0.050***	
	[0.017]	
Constant	-0.087***	
	[0.022]	
Journal fixed effects	**YES**	
Observations	96,543	
R-squared	0.068	

Robust standard errors in brackets
*** p<0.01, ** p<0.05, * p<0.1

Table S15. LPM of positive presentation in broader set of clinical articles

Sample of broader publication categories	
Article	69,360 (63%)
Conference paper	1,017 (1%)
Letter	29,325 (31%)
Survey	5,365 (6%)
	95,067
Linear Probability Model	
Dependent variable: Positive article = 1	
Independent variables	
Female first and last author = 1	-0.020***
	[0.004]
Percent female authors on byline	0.016***
	[0.004]
Count of authors on byline	0.005***
	[0.000]
Percentile positive MeSH field	0.002***
	[0.000]
Publication year fixed effects	
Fixed_effect_2003	0.002
	[0.005]
Fixed_effect_2004	0.002
	[0.005]
Fixed_effect_2005	-0.000
	[0.005]
Fixed_effect_2006	0.005
	[0.005]
Fixed_effect_2007	0.008
	[0.005]
Fixed_effect_2008	0.009*
	[0.005]
Fixed_effect_2009	0.019***
	[0.005]
Fixed_effect_2010	0.016***
	[0.005]
Fixed_effect_2011	0.015***
	[0.005]
Fixed_effect_2012	0.016***
	[0.005]
Fixed_effect_2013	0.016***
	[0.005]
Fixed_effect_2014	0.010**
	[0.005]
Fixed_effect_2015	0.016***
	[0.005]
Fixed_effect_2016	0.023***
	[0.005]
Fixed_effect_2017	0.043***
	[0.006]
Constant	-0.034***
	[0.005]
Journal fixed effects	**YES**
Observations	95,067
R-squared	0.118

Standard errors in brackets
*** p<0.01, ** p<0.05, * p<0.1

Table S16. LPM of positive presentation with citations as independent variable

Linear Probability Model	
Dependent variable: Positive article = 1	
Independent variables	
Female first and last author = 1	-0.011**
	[0.005]
Percent female authors on byline	-0.001
	[0.006]
Count of authors on byline	0.003***
	[0.000]
Forward citations (log)	0.016***
	[0.001]
Publication year fixed effects	
Fixed_effect_2003	0.001
	[0.007]
Fixed_effect_2004	0.003
	[0.007]
Fixed_effect_2005	0.008
	[0.007]
Fixed_effect_2006	0.017**
	[0.007]
Fixed_effect_2007	0.021***
	[0.007]
Fixed_effect_2008	0.024***
	[0.008]
Fixed_effect_2009	0.038***
	[0.008]
Fixed_effect_2010	0.038***
	[0.008]
Fixed_effect_2011	0.039***
	[0.008]
Fixed_effect_2012	0.042***
	[0.008]
Fixed_effect_2013	0.055***
	[0.008]
Fixed_effect_2014	0.050***
	[0.008]
Fixed_effect_2015	0.067***
	[0.008]
Fixed_effect_2016	0.070***
	[0.008]
Fixed_effect_2017	0.079***
	[0.008]
Constant	0.022***
	[0.007]
Journal fixed effects	**YES**
Observations	61,969
R-squared	0.095

Standard errors in brackets

*** p<0.01, ** p<0.05, * p<0.1

Table S17. Poisson regression of count of positive words

Poisson Model	
Dependent variable: Number of positive words	
Independent variables	
Female first and last author = 1	0.586***
	[0.028]
Percent female authors on byline	0.985
	[0.047]
Count of authors on byline	1.011***
	[0.001]
Percentile positive MeSH field	1.030***
	[0.000]
Publication year fixed effects	
Fixed_effect_2003	0.957
	[0.064]
Fixed_effect_2004	1.033
	[0.067]
Fixed_effect_2005	1.138**
	[0.071]
Fixed_effect_2006	1.096
	[0.070]
Fixed_effect_2007	1.150**
	[0.073]
Fixed_effect_2008	1.275***
	[0.080]
Fixed_effect_2009	1.401***
	[0.086]
Fixed_effect_2010	1.428***
	[0.091]
Fixed_effect_2011	1.394***
	[0.089]
Fixed_effect_2012	1.398***
	[0.090]
Fixed_effect_2013	1.500***
	[0.094]
Fixed_effect_2014	1.378***
	[0.087]
Fixed_effect_2015	1.502***
	[0.092]
Fixed_effect_2016	1.562***
	[0.102]
Fixed_effect_2017	1.727***
	[0.123]
Constant	0.029***
	[0.002]
Observations	59,360
Pseudo R-squared	0.076
Standard errors in brackets	
*** p<0.01, ** p<0.05, * p<0.1	

Note: Incidence rate ratios shown

Figure S18. Percent relative gender difference in positive presentation by gender composition for (A) clinical research articles and (B) PubMed articles

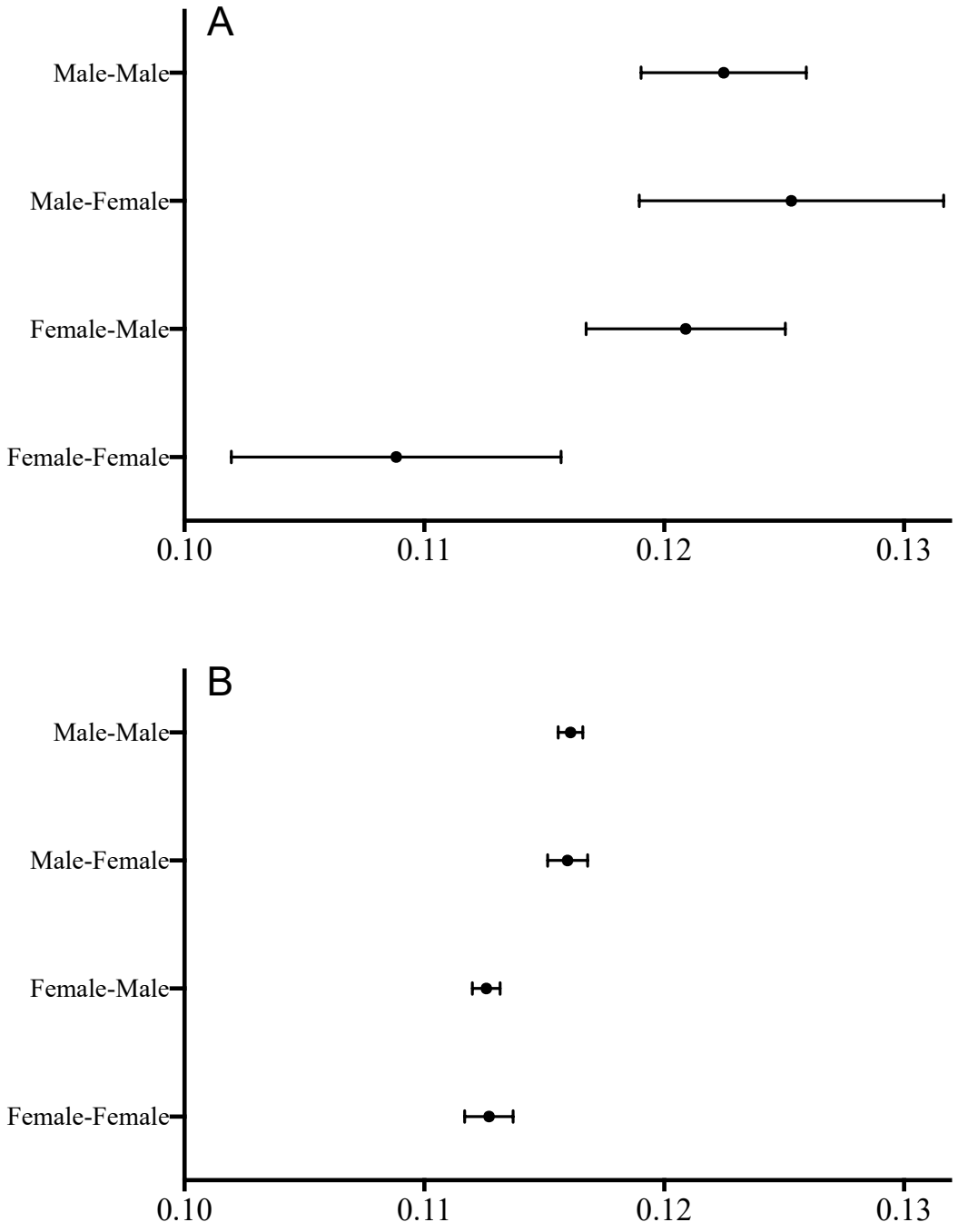


Table S19. Overview of estimated effects of positive presentation on subsequent citations

Effect of positive presentation on citations	Clinical articles (JIF >10)	Clinical articles (JIF ≤10)	PubMed articles (JIF >10)	PubMed articles (JIF ≤10)
OLS model (Citations)	13%***	3%	12%***	6%***
OLS model (Scopus citations)	38%***	16%***		
Poisson model (Citations)	19%***	13%***	30%***	15%***
Poisson model (Scopus citations)	20%***	14%***		

Table S20. OLS models of effect of positive presentation on subsequent citations

Models	(1)	(2)	(3)	(4)	(5)	(6)
	Clinical articles (all JIFs)	**Clinical articles (JIF >10)**	**Clinical articles (JIF ≤10)**	**PubMed articles (all JIFs)**	**PubMed articles (JIF >10)**	**PubMed articles (JIF ≤10)**
	OLS	**OLS**	**OLS**	**OLS**	**OLS**	**OLS**
Dependent variable	**Citations (log)**	**Citations (log)**	**Citations (log)**	**Citations (log)**	**Citations (log)**	**Citations (log)**
Independent variables						
Dummy for positive (1) article	1.094***	1.130***	1.033	1.060***	1.120***	1.057***
	[0.014]	[0.018]	[0.020]	[0.001]	[0.009]	[0.001]
Female first and last author = 1	0.847***	0.775***	0.953**	0.970***	0.761***	0.978***
	[0.012]	[0.015]	[0.020]	[0.002]	[0.009]	[0.002]
Percent female authors on byline	1.306***	1.384***	1.239***	1.093***	1.233***	1.088***
	[0.024]	[0.034]	[0.035]	[0.002]	[0.016]	[0.002]
Count of authors on byline	1.027***	1.027***	1.025***	1.020***	1.023***	1.018***
	[0.001]	[0.001]	[0.002]	[0.000]	[0.000]	[0.000]
Percentile positive MeSH field	1.003***	1.003***	1.003***	1.005***	1.014***	1.004***
	[0.000]	[0.000]	[0.001]	[0.000]	[0.000]	[0.000]
Publication year fixed effects						
Fixed_effect_2003	0.995	0.989	1.006	1.050***	1.023	1.053***
	[0.022]	[0.027]	[0.038]	[0.003]	[0.016]	[0.003]
Fixed_effect_2004	0.976	0.977	0.970	1.082***	1.040**	1.085***
	[0.022]	[0.028]	[0.036]	[0.003]	[0.017]	[0.003]
Fixed_effect_2005	0.995	0.983	1.022	1.128***	1.192***	1.125***
	[0.022]	[0.028]	[0.038]	[0.003]	[0.019]	[0.003]
Fixed_effect_2006	0.977	0.986	0.963	1.184***	1.338***	1.174***
	[0.022]	[0.029]	[0.036]	[0.003]	[0.021]	[0.003]
Fixed_effect_2007	0.967	0.928**	1.060	1.294***	1.484***	1.282***
	[0.022]	[0.027]	[0.040]	[0.004]	[0.024]	[0.004]
Fixed_effect_2008	1.812***	1.721***	1.966***	1.486***	1.921***	1.470***
	[0.043]	[0.053]	[0.073]	[0.004]	[0.034]	[0.004]
Fixed_effect_2009	2.484***	2.643***	2.259***	1.626***	3.054***	1.572***
	[0.059]	[0.082]	[0.086]	[0.005]	[0.056]	[0.005]
Fixed_effect_2010	2.420***	2.505***	2.280***	1.640***	3.256***	1.578***
	[0.060]	[0.080]	[0.091]	[0.005]	[0.061]	[0.005]
Fixed_effect_2011	2.466***	2.545***	2.341***	1.684***	3.487***	1.617***
	[0.061]	[0.081]	[0.093]	[0.005]	[0.065]	[0.005]
Fixed_effect_2012	2.532***	2.651***	2.365***	1.717***	3.588***	1.648***
	[0.062]	[0.084]	[0.093]	[0.005]	[0.066]	[0.005]
Fixed_effect_2013	2.599***	2.724***	2.468***	1.734***	3.266***	1.672***
	[0.064]	[0.087]	[0.095]	[0.005]	[0.060]	[0.005]
Fixed_effect_2014	2.473***	2.300***	2.718***	1.772***	3.192***	1.712***
	[0.059]	[0.071]	[0.103]	[0.005]	[0.058]	[0.005]
Fixed_effect_2015	2.518***	2.382***	2.709***	1.772***	3.464***	1.705***
	[0.060]	[0.074]	[0.102]	[0.005]	[0.064]	[0.005]
Fixed_effect_2016	2.393***	2.157***	2.729***	1.772***	3.672***	1.701***
	[0.060]	[0.071]	[0.108]	[0.006]	[0.071]	[0.005]
Fixed_effect_2017	2.259***	2.148***	2.417***	1.671***	3.124***	1.608***
	[0.065]	[0.084]	[0.106]	[0.006]	[0.070]	[0.006]
Constant	0.870***	0.898***	0.823***	1.177***	0.631***	1.227***
	[0.019]	[0.024]	[0.029]	[0.004]	[0.012]	[0.004]
Journal fixed effects	**YES**	**YES**	**YES**	**YES**	**YES**	**YES**
Observations	96,543	59,360	37,183	5,046,315	237,783	4,808,532
R-squared	0.253	0.230	0.297	0.635	0.492	0.644

Standard errors in brackets; exponentiated coefficients shown

*** p<0.01, ** p<0.05, * p<0.1

Table S21. OLS models of effect of positive presentation on citations including gender interactions in high-impact journals

Models	(1)	(2)
	Clinical articles (JIF >10)	**PubMed articles (JIF >10)**
Dependent variable	**Citations (log)**	**Citations (log)**
Independent variables		
Dummy for positive (1) article = 1	1.087***	1.102***
	[0.019]	[0.009]
Female first and last author = 1	0.739***	0.740***
	[0.015]	[0.009]
Positive article X female first and last author	**1.517*** **	**1.213*** **
	[0.081]	[0.034]
Percent female authors on byline	1.419***	1.239***
	[0.035]	[0.016]
Count of authors on byline	1.027***	1.023***
	[0.001]	[0.000]
Percentile positive MeSH field	1.003***	1.014***
	[0.000]	[0.000]
Publication year fixed effects		
Fixed_effect_2003	0.988	1.023
	[0.027]	[0.016]
Fixed_effect_2004	0.977	1.040**
	[0.028]	[0.017]
Fixed_effect_2005	0.983	1.192***
	[0.028]	[0.019]
Fixed_effect_2006	0.987	1.338***
	[0.029]	[0.021]
Fixed_effect_2007	0.927**	1.484***
	[0.027]	[0.024]
Fixed_effect_2008	1.720***	1.920***
	[0.053]	[0.034]
Fixed_effect_2009	2.636***	3.053***
	[0.081]	[0.056]
Fixed_effect_2010	2.500***	3.256***
	[0.079]	[0.061]
Fixed_effect_2011	2.540***	3.487***
	[0.081]	[0.065]
Fixed_effect_2012	2.647***	3.588***
	[0.084]	[0.066]
Fixed_effect_2013	2.718***	3.267***
	[0.087]	[0.060]
Fixed_effect_2014	2.295***	3.193***
	[0.071]	[0.058]
Fixed_effect_2015	2.378***	3.465***
	[0.074]	[0.064]
Fixed_effect_2016	2.151***	3.672***
	[0.071]	[0.071]
Fixed_effect_2017	2.143***	3.123***
	[0.083]	[0.070]
Constant	0.900***	0.632***
	[0.024]	[0.012]
Journal fixed effects	**YES**	**YES**
Observations	59,360	237,783
R-squared	0.231	0.492

Standard errors in brackets; exponentiated coefficients shown

*** p<0.01, ** p<0.05, * p<0.1

Table S22. Poisson models of effect of positive presentation on citations

Models	(1)	(2)	(3)	(4)
	Clinical articles (JIF >10)	**Clinical articles (JIF ≤10)**	**PubMed articles (JIF >10)**	**PubMed articles (JIF ≤10)**
	Poisson	**Poisson**	**Poisson**	**Poisson**
Dependent variable	**Citations**	**Citations**	**Citations**	**Citations**
Independent variables				
Dummy for positive (1) article	1.194***	1.132***	1.300***	1.151***
	[0.027]	[0.041]	[0.015]	[0.003]
Female first and last author = 1	0.726***	1.037	0.762***	0.859***
	[0.024]	[0.036]	[0.013]	[0.003]
Percent female authors on byline	1.813***	1.307***	1.544***	1.410***
	[0.072]	[0.069]	[0.030]	[0.007]
Count of authors on byline	1.008***	1.021***	1.006***	1.007***
	[0.001]	[0.002]	[0.000]	[0.001]
Percentile positive MeSH field	1.010***	1.005***	1.016***	1.020***
	[0.000]	[0.001]	[0.000]	[0.000]
Publication year fixed effects				
Fixed_effect_2003	0.830	0.951	1.122***	1.164***
	[0.100]	[0.187]	[0.045]	[0.013]
Fixed_effect_2004	1.036	0.697*	1.146***	1.264***
	[0.117]	[0.145]	[0.045]	[0.014]
Fixed_effect_2005	1.448***	1.386*	1.297***	1.433***
	[0.162]	[0.260]	[0.048]	[0.015]
Fixed_effect_2006	1.608***	1.378*	1.573***	1.602***
	[0.173]	[0.254]	[0.056]	[0.016]
Fixed_effect_2007	2.770***	2.417***	1.852***	2.021***
	[0.272]	[0.402]	[0.061]	[0.020]
Fixed_effect_2008	8.506***	8.969***	2.743***	2.737***
	[0.753]	[1.329]	[0.090]	[0.026]
Fixed_effect_2009	12.839***	9.499***	3.504***	3.267***
	[1.107]	[1.395]	[0.109]	[0.030]
Fixed_effect_2010	12.728***	10.233***	3.625***	3.554***
	[1.105]	[1.502]	[0.114]	[0.032]
Fixed_effect_2011	13.018***	11.772***	3.898***	3.786***
	[1.130]	[1.723]	[0.120]	[0.034]
Fixed_effect_2012	14.259***	11.198***	3.931***	4.064***
	[1.243]	[1.645]	[0.120]	[0.037]
Fixed_effect_2013	14.701***	11.089***	3.857***	4.271***
	[1.272]	[1.620]	[0.120]	[0.038]
Fixed_effect_2014	12.606***	13.271***	4.009***	4.574***
	[1.096]	[1.923]	[0.122]	[0.041]
Fixed_effect_2015	13.160***	13.994***	4.203***	4.595***
	[1.144]	[2.028]	[0.130]	[0.041]
Fixed_effect_2016	12.421***	14.583***	4.900***	4.664***
	[1.099]	[2.120]	[0.152]	[0.042]
Fixed_effect_2017	12.198***	14.309***	5.533***	4.610***
	[1.124]	[2.089]	[0.185]	[0.046]
Constant	0.487***	0.502***	1.876***	0.977**
	[0.041]	[0.074]	[0.059]	[0.009]
Journal fixed effects	**NO**	**NO**	**NO**	**NO**
Observations	59,360	37,183	237,783	4,808,532

Standard errors in brackets; incidence rate ratios shown

*** p<0.01, ** p<0.05, * p<0.1

Table S23. OLS and Poisson model of effect of positive presentation on citations sourced from Scopus

Models	(1)	(2)	(3)	(4)
	Clinical articles (JIF >10)	**Clinical articles (JIF ≤10)**	**Clinical articles (JIF >10)**	**Clinical articles (JIF ≤10)**
	OLS	**OLS**	**Poisson**	**Poisson**
Dependent variable	**Scopus citations (log)**	**Scopus citations (log)**	**Scopus citations**	**Scopus citations**
Independent variables				
Dummy for positive (1) article	1.377***	1.160***	1.204***	1.136***
	[0.026]	[0.021]	[0.028]	[0.029]
Female first and last author = 1	0.423***	0.844***	0.483***	0.851***
	[0.010]	[0.016]	[0.021]	[0.021]
Percent female authors on byline	1.543***	1.178***	1.209***	1.020
	[0.043]	[0.030]	[0.048]	[0.035]
Count of authors on byline	1.075***	1.065***	1.013***	1.049***
	[0.001]	[0.001]	[0.001]	[0.002]
Percentile positive MeSH field	0.998***	0.997***	0.996***	0.997***
	[0.000]	[0.000]	[0.001]	[0.001]
Publication year fixed effects				
Fixed_effect_2003	0.883***	0.936**	0.887***	0.902**
	[0.028]	[0.032]	[0.038]	[0.041]
Fixed_effect_2004	0.871***	0.891***	0.870***	0.883***
	[0.028]	[0.030]	[0.039]	[0.042]
Fixed_effect_2005	0.866***	0.853***	0.831***	0.872***
	[0.028]	[0.029]	[0.033]	[0.041]
Fixed_effect_2006	0.811***	0.807***	0.842***	0.799***
	[0.027]	[0.027]	[0.037]	[0.035]
Fixed_effect_2007	0.722***	0.806***	0.714***	0.776***
	[0.024]	[0.028]	[0.029]	[0.035]
Fixed_effect_2008	0.858***	0.769***	0.798***	0.730***
	[0.030]	[0.026]	[0.034]	[0.031]
Fixed_effect_2009	0.851***	0.718***	0.816***	0.649***
	[0.030]	[0.025]	[0.036]	[0.030]
Fixed_effect_2010	0.744***	0.681***	0.774***	0.600***
	[0.027]	[0.025]	[0.038]	[0.026]
Fixed_effect_2011	0.605***	0.591***	0.592***	0.540***
	[0.022]	[0.021]	[0.025]	[0.024]
Fixed_effect_2012	0.558***	0.528***	0.550***	0.443***
	[0.020]	[0.019]	[0.032]	[0.019]
Fixed_effect_2013	0.503***	0.438***	0.434***	0.396***
	[0.018]	[0.015]	[0.018]	[0.016]
Fixed_effect_2014	0.440***	0.346***	0.409***	0.308***
	[0.016]	[0.012]	[0.026]	[0.013]
Fixed_effect_2015	0.375***	0.295***	0.349***	0.257***
	[0.013]	[0.010]	[0.024]	[0.011]
Fixed_effect_2016	0.320***	0.241***	0.296***	0.185***
	[0.012]	[0.009]	[0.015]	[0.009]
Fixed_effect_2017	0.244***	0.175***	0.210***	0.123***
	[0.011]	[0.007]	[0.010]	[0.006]
Constant	28.435***	27.373***	149.230***	59.948***
	[0.881]	[0.880]	[5.726]	[2.557]
Journal fixed effects	**YES**	**YES**	**NO**	**NO**
Observations	59,356	37,177	59,356	37,177
R-squared	0.345	0.337	0.123	0.182

Standard errors in brackets; exponentiated coefficients (OLS) and incidence rate ratios (Poisson) shown; Pseudo R-squared shown for Poisson models

*** p<0.01, ** p<0.05, * p<0.1

Table S24. Gender differences in positive presentation with different gender designation thresholds

Models		(1)	(2)	(3)	(4)
		Clinical articles (JIF >10)	**Clinical articles (JIF ≤10)**	**PubMed articles (JIF >10)**	**PubMed articles (JIF ≤10)**
90% gender designation threshold	Female first and last author = 1	10.7%	11.0%	12.8%	11.3%
	Female first and last author = 0	12.9%	11.2%	14.4%	11.3%
	Percent relative difference	***21.4%***	***1.4%***	***12.8%***	***0.1%***
95% gender designation threshold	Female first and last author = 1	10.6%	10.7%	12.6%	11.3%
	Female first and last author = 0	12.9%	11.1%	14.3%	11.3%
	Percent relative difference	***21.3%***	***3.1%***	***13.1%***	***0.1%***
80% gender designation threshold	Female first and last author = 1	10.8%	10.8%	12.9%	11.4%
	Female first and last author = 0	13.1%	11.3%	14.5%	11.4%
	Percent relative difference	***21.5%***	***4.1%***	***12.8%***	***0.0%***
50% gender designation threshold	Female first and last author = 1	11.7%	10.9%	13.5%	11.6%
	Female first and last author = 0	13.4%	11.4%	14.9%	11.7%
	Percent relative difference	***14.6%***	***4.4%***	***10.3%***	***0.7%***

Table S25. Comparison of articles with and without gender designation

Characteristics	Non-gender designated articles	Gender designated articles at 90% probability
JIF > 10 (%)	5.2%	7.4%
JIF ≤ 10(%)	94.8%	92.6%
Number of authors	5.98	4.64
Median publication year	2012	2011
N	3,893,491	6,793,935

Table S26. Absolute and relative gender differences in individual word use for top 10 positive words in clinical research articles

Individual word	Representation (%)	Female first and last authors = 1	Female first and last authors = 0	Absolute difference	Lower CI 95%	Upper CI 95%	Relative Difference	Lower CI 95%	Upper CI 95%
novel	44.82	3.6%	5.8%	-2.145%	-2.5%	-1.8%	59.2%	69.7%	48.8%
unique	13.30	1.2%	1.7%	-0.515%	-0.7%	-0.3%	43.8%	61.7%	25.9%
promising	12.54	0.9%	1.6%	-0.682%	-0.9%	-0.5%	72.3%	94.0%	50.6%
favorable	12.31	0.8%	1.6%	-0.818%	-1.0%	-0.6%	101.8%	127.1%	76.6%
robust	6.68	0.5%	0.9%	-0.349%	-0.5%	-0.2%	67.8%	97.0%	38.7%
excellent	6.53	0.5%	0.9%	-0.394%	-0.5%	-0.2%	85.7%	117.9%	53.4%
prominent	4.49	0.3%	0.6%	-0.231%	-0.4%	-0.1%	66.2%	101.5%	31.0%
supportive	2.78	0.4%	0.3%	0.045%	-0.1%	0.1%	-12.0%	13.9%	-37.9%
encouraging	2.64	0.4%	0.3%	-0.169%	-0.3%	-0.1%	45.2%	70.4%	19.9%
remarkable	1.58	0.1%	0.2%	-0.074%	-0.1%	0.0%	57.4%	114.1%	0.6%

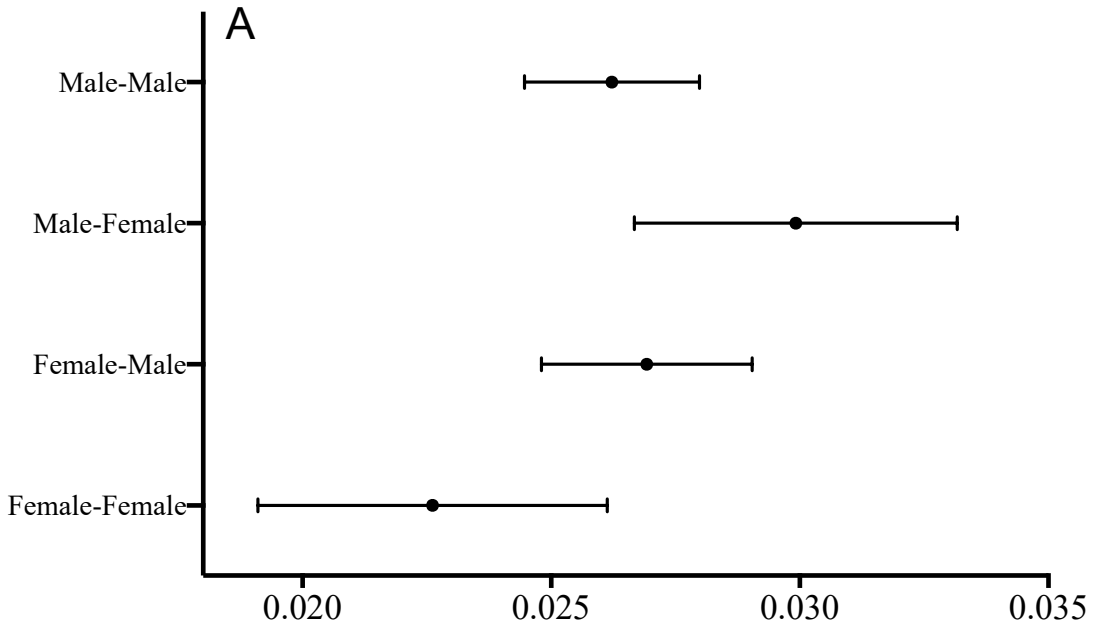
Table S27. Illustration of MeSH-based field control

BMJ. 2016 Jan 19;532:h6895. doi: 10.1136/bmj.h6895.

Patient choice in opt-in, active choice, and opt-out HIV screening: randomized clinical trial.

MeSH terms	MeSH ID	Percentile association with positive presentation
Adolescent	D000293	23.34
Adult	D000328	23.50
Choice Behavior*	D002755	45.24
Emergency Service, Hospital	D004636	7.34
Female	D005260	29.67
HIV Infections/diagnosis*	D006784	7.08
Hospitals, Teaching	D006786	8.13
Hospitals, Urban	D006801	43.06
Humans	D008297	31.03
Male	D008403	13.68
Mass Screening/statistics & numerical data*	D008875	19.09
Middle Aged	D010342	13.32
Patient Acceptance of Health Care*	D015658	35.81
Patient Preference*	D055815	27.25
Young Adult	D057240	9.99
Median		23.34

Figure S28. Percent relative gender difference in the use of negative words by gender composition for (A) clinical research articles and (B) PubMed articles



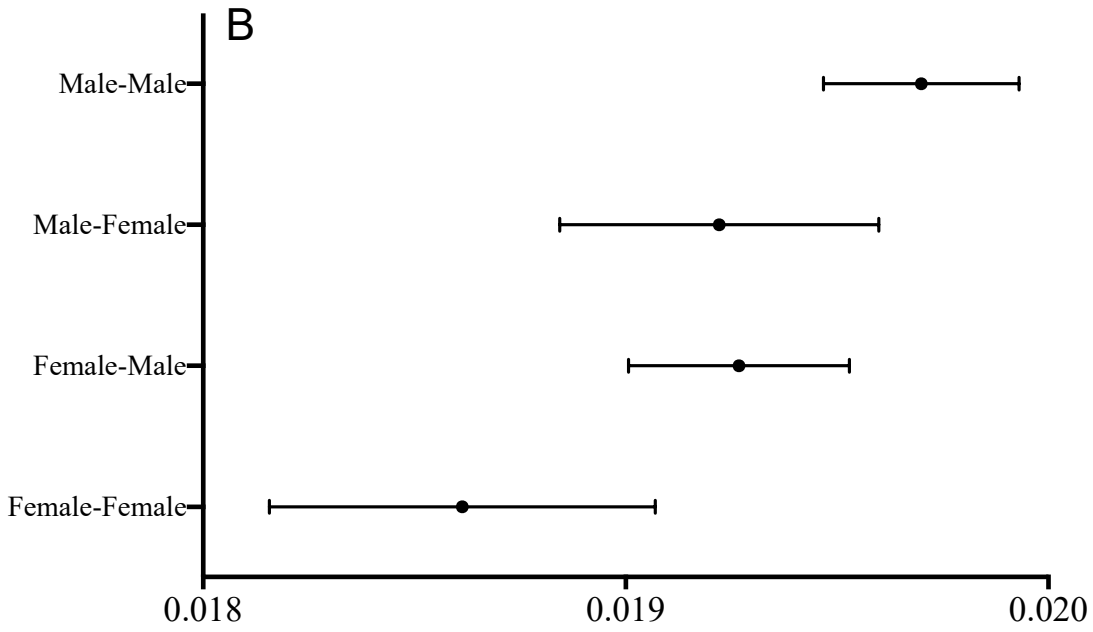


Table S29. Absolute effect and relative effect differences and 95% confidence intervals

	Percent of positive presented articles							
Samples	Female first and last author (1)	Female first and last author (0)	Absolute difference	Lower CI	Upper CI	Relative Difference	Lower CI	Upper CI
Clinical articles (JIF > 10)	10.7%	12.9%	-2.3%	-3.3%	-1.3%	21.4%	30.5%	12.3%
Clinical articles (JIF ≤ 10)	11.0%	11.2%	-0.2%	-1.2%	0.9%	1.4%	11.2%	-8.4%
PubMed articles (JIF > 10)	12.8%	14.4%	-1.6%	-2.2%	-1.0%	12.8%	17.4%	8.2%
PubMed articles (JIF ≤ 10)	11.3%	11.3%	0.0%	-0.1%	0.1%	0.1%	1.0%	-0.8%

Table S30. Linear probability model of probability of positive presentation with MeSH-based field matching

	Proportion positive articles
Predicted margins	**Clinical articles (JIF > 10)**
Female first and last author = 1	9.4%
Female first and last author = 0	11.7%
Linear Probability Model (WLS)	
Dependent variable: Positive article = 1	
Independent variables	
Female first and last author = 1	-0.023***
	[0.005]
Percent female authors on byline	0.002
	[0.006]
Count of authors on byline	0.004***
	[0.000]
fe_2003	-0.002
	[0.007]
fe_2004	0.001
	[0.007]
fe_2005	0.002
	[0.007]
fe_2006	0.011
	[0.007]
fe_2007	0.015**
	[0.007]
fe_2008	0.020***
	[0.007]
fe_2009	0.032***
	[0.007]
fe_2010	0.030***
	[0.008]
fe_2011	0.027***
	[0.008]
fe_2012	0.030***
	[0.008]
fe_2013	0.039***
	[0.008]
fe_2014	0.035***
	[0.007]
fe_2015	0.045***
	[0.007]
fe_2016	0.051***
	[0.008]
fe_2017	0.053***
	[0.009]
Constant	0.082
	[0.317]
Journal fixed effects	**YES**
Match strata fixed effects	**YES**
Observations	61,973
R-squared	0.101

Standard errors in brackets
*** p<0.01, ** p<0.05, * p<0.1

Weighted least squares regression, including weights for 99 matched strata based on MeSH term percentile ranking.

Table S31. Linear probability model of probability of positive presentation in articles' introduction

	Proportion positive articles	
Predicted margins		
Female first and last author = 1	31.0%	27.1%
Female first and last author = 0	21.0%	19.3%
Linear Probability Model		
Dependent variable: Positive article = 1		
Independent variables		
Female first and last author = 1	0.100	0.078
	[0.196]	[0.197]
Percent female authors on byline	0.088	0.117
	[0.305]	[0.301]
Count of authors on byline	-0.025	-0.030
	[0.023]	[0.022]
Percentile introduction length	0.008*	0.009**
	[0.004]	[0.004]
Percentile positive MeSH field		0.008
		[0.007]
fe_2007	-0.520	-0.465
	[0.604]	[0.596]
fe_2008	-1.227**	-1.190*
	[0.565]	[0.565]
fe_2009	-1.338**	-1.277**
	[0.526]	[0.524]
fe_2010	-0.946*	-0.979*
	[0.486]	[0.481]
fe_2011	-1.085*	-0.954*
	[0.536]	[0.538]
fe_2012	-0.745	-0.643
	[0.474]	[0.473]
fe_2013	-1.221**	-1.127**
	[0.521]	[0.516]
fe_2014	-1.042*	-0.999*
	[0.501]	[0.503]
fe_2015	-0.829	-0.785
	[0.482]	[0.478]
fe_2016	-0.846	-0.894
	[0.519]	[0.531]
fe_2017	-1.022	-
	[0.626]	
Constant	0.930*	0.520
	[0.521]	[0.625]
Observations	100	86
R-squared	0.860	0.861

Standard errors in brackets
*** p<0.01, ** p<0.05, * p<0.1

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